

NEUROTUTORSIM

A Fully Computational Study of Traditional and AI-Mediated Education

Project brief and execution protocol for two Master in Business Analytics students

Project premise: This is a purely computational project. We will not recruit participants, collect neuroimaging data, or run an educational experiment. We will compare matched instructional regimes by combining generated educational materials, pretrained human-brain encoding models, synthetic learner simulations, public validation data, and explicit longitudinal assumptions. The ten-year results are scenario projections, not causal forecasts.

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1. Project overview

The objective of this project is to build a computational laboratory that compares how human learners might be affected by three educational regimes: **(1) traditional fixed instruction, (2) adaptive AI scaffolding, and (3) AI substitution or cognitive offloading**. We will model two layers separately. First, Meta s TRIBE v2 will estimate the immediate cortical response that an average adult human is predicted to show while receiving each instructional stimulus. Second, a synthetic learning and plasticity model will accumulate knowledge, effort, retrieval, forgetting, and neural-state updates across repeated learning episodes. The study will then simulate one-, five-, and ten-year trajectories under alternative assumptions.

Non-negotiable interpretation: TRIBE predicts stimulus-evoked brain activity. It does not predict long-term learning or plasticity. Any cumulative brain trajectory must therefore be generated by a separate model whose assumptions are explicit, varied, and stress-tested. We will never describe the ten-year outputs as observed or causally identified effects.

1.1 Primary research question

How do traditional fixed instruction, adaptive AI scaffolding, and AI substitution differ in their predicted immediate cortical engagement, simulated learning outcomes, and long-term functional neural trajectories under alternative models of human learning and plasticity?

1.2 Secondary research questions

- Which differences between traditional and AI materials remain after matching factual content, length, readability, modality, and presentation time?
- Does adaptive scaffolding produce a different predicted cortical profile from AI systems that provide complete solutions?
- Under which assumptions does AI improve independent retention and transfer, and under which assumptions does it create persistent dependence on support?
- How sensitive are long-term conclusions to plasticity rates, forgetting, exposure intensity, and the mapping from immediate activation to learning?
- Can we identify a robust policy frontier indicating when AI-mediated education is beneficial, neutral, or harmful relative to traditional instruction?

1.3 Expected contribution

The project will contribute a reproducible framework rather than a definitive empirical verdict. Its value lies in integrating a human-brain foundation model with educational stimulus engineering, synthetic learner dynamics, uncertainty propagation, and long-horizon scenario analysis. The central output will be a phase diagram showing the combinations of personalization, learner effort, answer substitution, fading support, and plasticity under which AI-mediated education outperforms or underperforms traditional materials.

1.4 Required deliverables

Deliverable	Minimum content	Owner
D1. Reproducible repository	Environment files, scripts, configuration files, data dictionaries, logs, tests, and a complete README.	Both

Deliverable	Minimum content	Owner
D2. Matched educational corpus	At least 120 learning units in two domains and three instructional regimes.	Student 1
D3. TRIBE prediction dataset	Time-resolved cortical predictions, parcel summaries, network metrics, and quality checks for every stimulus.	Student 2
D4. Synthetic learner engine	Knowledge tracing, effort, dependence, retrieval, confidence, and support withdrawal.	Student 1
D5. Longitudinal simulator	One-, five-, and ten-year Monte Carlo trajectories under multiple plasticity models.	Student 2
D6. Validation and robustness report	Negative controls, placebo tests, specification curves, and uncertainty decomposition.	Both
D7. Final paper and presentation	A journal-style manuscript, technical appendix, figures, and a 15-minute presentation.	Both

2. Short literature background

2.1 Foundation models of human cognition

Centaur demonstrates that a large language model can be adapted to predict human behavioural choices across a wide range of psychological tasks. It was obtained by fine-tuning a language model on Psych-101, a large corpus of trial-level human behaviour. The relevance for our project is methodological: synthetic learners should not be treated as generic chatbots. Where computationally feasible, they should be based on a model calibrated to human behavioural regularities. Centaur or the smaller Minitaur can therefore be used as one behavioural engine, although neither was trained specifically on university learning or ten-year educational development.

2.2 Foundation models of neural response

TRIBE v2 is a tri-modal foundation model designed to predict human fMRI responses to video, audio, and language. The model was trained on a unified dataset exceeding 1,000 hours of fMRI from 720 participants and is intended to generalize across novel stimuli, tasks, and subjects. For this project, TRIBE v2 will be used as an encoding model: educational stimuli are inputs and predicted cortical responses are outputs. Its released predictions should be interpreted as model-based responses for an average adult subject, not as measurements from a specific learner.

2.3 Shared neural cores and lightweight readouts

The mouse visual-cortex foundation model developed by Wang and colleagues provides a useful architectural principle. The model separates a high-capacity shared core from subject-specific readouts. The shared core learned representations that generalized across animals, while small readouts could be adapted with limited additional data. We will borrow this principle conceptually: the pretrained TRIBE encoder remains frozen, while our project computes task-, condition-, and scenario-specific summaries outside the model. We will not claim that results from mouse visual cortex directly establish human educational plasticity.

2.4 Learning, scaffolding, and cognitive offloading

The educational comparison is organized around two opposing mechanisms. Adaptive scaffolding may improve learning by matching explanations and difficulty to the learner's current state, correcting misconceptions, and gradually removing assistance. Cognitive offloading may improve assisted performance while reducing retrieval, generative reasoning, and productive struggle. The project therefore separates AI scaffolding from AI substitution. Treating all AI use as a single condition would hide the mechanism that matters most: how much cognitive work remains with the learner.

2.5 What this study can and cannot establish

Permissible claim	Impermissible claim
TRIBE predicts different immediate cortical responses to matched educational stimuli.	Students exposed to AI will necessarily exhibit those brain responses.
Under specified learning and plasticity assumptions, simulated trajectories diverge.	The model discovers the true brain effects of ten years of AI education.
Some conclusions are robust across a wide range of assumptions.	Robust simulation is equivalent to causal identification.
The framework identifies parameters requiring future empirical measurement.	The project replaces future educational or neuroscientific experiments.

3. Study design

3.1 Unit of analysis

The basic unit is an instructional episode. An episode consists of a problem, the learner's first response, an instructional intervention, a second response, and a transfer question. The same underlying knowledge objective must appear in every condition. The episode identifier must remain constant across the traditional, AI-scaffolding, and AI-substitution versions.

3.2 Domains and curriculum

The core implementation will use two quantitative domains that are appropriate for university students and allow deterministic validation:

- Bayesian probability and statistical reasoning.
- Causal inference and experimental design.

Each domain will contain 60 learning units, producing 120 matched units in total. Each unit must include one core concept, one worked problem, one practice problem, and one structurally novel transfer problem. The material should be pitched at students with basic quantitative literacy but no specialized prior training in the exact topic.

3.3 Instructional conditions

Condition	Adaptation	Cognitive work retained by learner	Operational rule
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Traditional fixed instruction	None	Moderate to high	Fixed explanation, worked example, fixed hint ladder, and prewritten feedback.
AI scaffolding	High	High	The tutor diagnoses the error, asks a guiding question, offers graduated hints, and withholds the final answer until the learner attempts the solution.
AI substitution	High	Low	The tutor provides the reasoning steps and final solution rapidly, minimizing required retrieval and generation.
Optional static AI control	None	Moderate	AI-generated material is presented without interaction and matched to the traditional version.

3.4 Experimental factors in the computational design

Although no human experiment is conducted, the computational design is factorial. The following variables must be independently manipulable in configuration files:

- Instructional regime: traditional, AI scaffolding, AI substitution, optional static AI.
- Domain: Bayesian reasoning or causal inference.
- Difficulty: introductory, intermediate, advanced.
- Learner prior knowledge: low, medium, high.
- Learner learning rate: slow, median, fast.
- AI support persistence: immediate withdrawal, gradual fading, persistent support.
- Exposure: low, medium, high sessions per week.
- Plasticity mechanism: activation accumulation, prediction-error learning, effort-dependent learning, hybrid.
- Forgetting: weak, moderate, strong.
- Noise: deterministic, low stochasticity, high stochasticity.

4. Data architecture

4.1 Repository structure

```
neurotutorsim/
  config/
  data/raw/
  data/interim/
  data/processed/
  stimuli/traditional/
  stimuli/ai_scaffolding/
  stimuli/ai_substitution/
  src/generation/
  src/validation/
  src/tribe/
  src/learners/
```

src/plasticity/
src/analysis/
src/visualization/
tests/
notebooks/
outputs/tables/
outputs/figures/
reports/

4.2 Core tables

Table	One row per	Required fields
units.csv	knowledge unit	unit_id, domain, concept, difficulty, prerequisites, reference_answer, transfer_answer
stimuli.csv	condition-specific stimulus	stimulus_id, unit_id, condition, text, modality, duration, word_count, readability, equation_count, example_count
responses.csv	synthetic learner turn	learner_id, episode, condition, prompt, response, answer, confidence, correctness, latency_proxy, token_count
tribe_vertex.parquet	stimulus-time-vertex prediction	stimulus_id, time_index, vertex_id, predicted_bold
tribe_parcel.parquet	stimulus-time-parcel summary	stimulus_id, time_index, parcel_id, network, mean_bold, sd_bold
learner_state.parquet	learner after each episode	learner_id, time, K, M, R, C, D, effort, support, correctness
neural_state.parquet	accumulated neural state	learner_id, time, parcel_id, state_value, mechanism, parameter_draw
simulation_draws.parquet	scenario-level outcomes	draw_id, scenario, year, outcome, estimate, parameter values

4.3 Reproducibility requirements

- Pin all package versions with conda-lock, uv.lock, Poetry, or an equivalent environment file.
- Use one master random seed and derived seeds for corpus generation, learner generation, and Monte Carlo draws.
- Write all prompts, model names, temperatures, decoding parameters, and API or checkpoint versions to machine-readable logs.
- Cache all generated materials and predictions. Never silently regenerate an existing item.
- Every figure and table must be generated from a script, not manually edited.
- Add unit tests for scoring, state updates, matching, and scenario aggregation.

5. Phase I: build and validate the educational corpus

5.1 Construct the reference curriculum

1. Create a concept map for each domain containing 20 concepts and directed prerequisite links.
2. Create three units per concept: introductory, intermediate, and advanced. This yields 60 units per domain.
3. For every unit, write a canonical problem, fully worked reference solution, common misconception, one near-transfer problem, and one far-transfer problem.
4. Validate all numeric answers with deterministic Python or R functions. Store the validator name and expected output.
5. Assign each unit a target completion time and a difficulty score from 1 to 5.

5.2 Generate the three instructional versions

The traditional version should be written first and treated as the content anchor. AI versions must preserve the same factual target. They may differ in adaptation and sequencing, not in the truth of the underlying material.

6. Traditional: produce a fixed explanation of 250-400 words, one worked example, and a three-level prewritten hint ladder.
7. AI scaffolding: create a tutor policy prompt that diagnoses the learner's error, asks one question at a time, and releases hints from general to specific. The final answer is withheld until the learner attempts the problem or exhausts the hint ladder.
8. AI substitution: create a tutor policy prompt that provides the complete explanation, calculation, and answer immediately after the first error.
9. Generate at least three stylistic variants of every stimulus. Retain one primary version and two robustness versions.
10. Convert all retained stimuli to standardized text and, if computationally feasible, standardized audio using the same voice, rate, and prosody.

5.3 Match the materials

The three conditions must be matched as closely as possible on non-pedagogical features. For every stimulus s , construct a vector of observable properties:

$$x_s = [\text{words}, \text{characters}, \text{sentences}, \text{reading level}, \text{equations}, \text{examples}, \text{duration}, \text{lexical diversity}, \text{semantic coverage}] \quad (1)$$

For each traditional stimulus, select or edit the AI versions to minimize standardized Mahalanobis distance:

$$d(s_T, s_A) = \sqrt{(x_T - x_A)^T S^{-1} (x_T - x_A)} \quad (2)$$

where S is the pooled covariance matrix of matching features. Exact matching should be imposed on unit_id, domain, concept, difficulty, modality, and correctness. Caliper matching should be imposed on duration and word count. The target standardized mean difference for every continuous matching feature is $|SMD| < 0.10$.

$$SMD_k = (\text{mean}(x_{k,T}) - \text{mean}(x_{k,A})) / \sqrt{(\text{var}(x_{k,T}) + \text{var}(x_{k,A}))/2} \quad (3)$$

5.4 Automated content validation

- Correctness: execute numeric answers or compare symbolic expressions using SymPy.
- Coverage: calculate semantic similarity between the condition-specific explanation and the canonical reference solution.
- Contradiction: use a separate judge model to identify factual contradictions, missing assumptions, or invalid causal statements.
- Readability: calculate Flesch-Kincaid grade, sentence length, word length, and syntactic depth.
- Pedagogical-policy compliance: classify whether the response asks a guiding question, provides a hint, or reveals the answer.
- Leakage: detect whether scaffolding material exposes the final answer before the required learner attempt.

5.5 Corpus acceptance criteria

Criterion	Threshold	Action if failed
Reference correctness	100%	Correct manually and rerun all validators.
Condition correctness	At least 99%; no retained critical errors	Regenerate or edit failed stimuli.
Semantic coverage	Cosine similarity above preregistered threshold; inspect lower tail	Review lowest 10% manually.
Matching	All SMD < 0.10; duration caliper within 10%	Edit or replace versions.
Scaffolding leakage	Below 2%	Tighten tutor policy and regenerate.
Duplicate content	No near duplicates across units	Remove or rewrite duplicates.

6. Phase II: generate predicted cortical responses with TRIBE v2

6.1 Environment and model verification

11. Clone the official TRIBE v2 repository and install the exact environment specified by the authors.
12. Download the official released checkpoint and verify its checksum.
13. Run the supplied inference notebook on the official example stimuli.
14. Reproduce at least one provided visualization or summary before processing our data.
15. Record checkpoint, commit hash, hardware, precision, batch size, and runtime.

6.2 Stimulus timing

TRIBE is time-sensitive. We therefore need a deterministic mapping from content to presentation time. For text, use a fixed reading speed r words per minute and preserve sentence boundaries. The onset of word j is:

$$onset_j = 60 * cumulative_words_before_j / r \quad (4)$$

Use $r = 220$ words per minute in the main specification, with 180 and 260 words per minute as robustness values. For audio, use identical text-to-speech settings across conditions and retain word-level timestamps.

6.3 Inference output

For stimulus s , TRIBE yields a predicted response matrix:

$$B_{\hat{s}} \text{ in } R^{(T_s \times V)} \quad (5)$$

where T_s is the number of predicted time points and V is the number of cortical vertices. Store the complete output in compressed columnar format. Do not retain only static brain images.

6.4 Parcel and network aggregation

Map vertices to a standard cortical parcellation. For parcel p containing vertex set V_p , calculate:

$$B_{\hat{spt}} = (1 / |V_p|) * \sum_{v \in V_p} B_{\hat{stv}} \quad (6)$$

For network n containing parcel set P_n :

$$B_{\hat{snt}} = (\sum_{p \in P_n} w_p B_{\hat{spt}}) / (\sum_{p \in P_n} w_p) \quad (7)$$

where w_p is parcel surface area or number of vertices. Use area weighting in the main analysis and equal parcel weighting as a robustness check.

6.5 Immediate neural metrics

Calculate the following metrics for each stimulus, parcel, and network:

- Mean predicted response: average over the instructional window.
- Peak response: maximum predicted response after temporal smoothing.
- Area under the curve: trapezoidal integral across time.
- Time to peak: time from stimulus onset to maximum response.
- Sustained engagement: proportion of time points above a stimulus-specific baseline threshold.
- Spatial dispersion: variance of parcel responses.
- Spatial entropy: normalized entropy of non-negative parcel weights.
- Network integration: average pairwise coactivation or correlation among selected networks.
- Representational differentiation: distance between predicted patterns for distinct concepts.
- Within-concept consistency: similarity between variants of the same concept.

$$AUC_{sp} = \sum_t 0.5 * (B_{\hat{sp},t} + B_{\hat{sp},t+1}) * \Delta_t \quad (8)$$

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$$q_{sp} = \max(B_{\hat{sp}}, 0) / \sum_p \max(B_{\hat{sp}}, 0); H_s = -\sum_p q_{sp} \log(q_{sp}) / \log(P) \quad (9)$$

6.6 Condition contrasts

For each matched unit u and neural metric m , compute paired contrasts:

$$\Delta_{m,u}^{(S-T)} = m_{u,scaffolding} - m_{u,traditional} \quad (10)$$

$$\Delta_{m,u}^{(U-T)} = m_{u,substitution} - m_{u,traditional} \quad (11)$$

$$\Delta_{m,u}^{(S-U)} = m_{u,scaffolding} - m_{u,substitution} \quad (12)$$

Estimate average differences with unit fixed effects and clustered or bootstrap uncertainty:

$$m_{uc} = \alpha_u + \beta_1 I(scaffolding) + \beta_2 I(substitution) + \gamma^T x_{uc} + \epsilon_{uc} \quad (13)$$

Because these are deterministic model predictions conditional on generated materials, inferential intervals should primarily reflect stimulus sampling and model-generation uncertainty. Use a cluster bootstrap resampling units and regenerations, not naive vertex-level p-values.

6.7 Representational similarity analysis

For each condition c , summarize each unit by a parcel-pattern vector z_{uc} . Construct a representational dissimilarity matrix:

$$RDM_c[u, u'] = 1 - \text{corr}(z_{uc}, z_{u'c}) \quad (14)$$

Compare RDMs across conditions with Spearman correlation and permutation tests that shuffle condition labels within matched units. Assess whether AI scaffolding preserves conceptual geometry, increases separation between concepts, or homogenizes responses.

7. Phase III: construct the synthetic learner population

7.1 Learner state

Simulate at least 5,000 learners. Each learner i has a state vector at episode t :

$$s_{i,t} = [K_{i,t}, M_{i,t}, R_{i,t}, C_{i,t}, D_{i,t}]^T \quad (15)$$

The components are: K, domain knowledge; M, memory strength; R, independent reasoning capacity; C, confidence calibration; and D, dependence on external support. Each component is scaled to [0,1]. Initialize learners by drawing correlated values from a truncated multivariate normal or Gaussian copula.

7.2 Initial population distributions

Use three prior-knowledge strata with mixture weights 0.30, 0.50, and 0.20 for low, medium, and high knowledge. Draw individual learning and forgetting parameters:

$$\alpha_i \sim \text{LogNormal}(\mu_\alpha, \sigma_\alpha); \delta_i \sim \text{Beta}(a_\delta, b_\delta) \quad (16)$$

The baseline parameter values must be treated as assumptions. Define low, medium, and high settings and include all of them in the simulation grid.

7.3 Cognitive-model response generation

Use Minitaur or Centaur as one response engine if computational resources permit. Every model call must include the learner state translated into observable history rather than directly revealing latent variables. The prompt should contain prior answers, errors, hint usage, and confidence. The output must be parsed into answer, confidence, explanation, and requested support.

In parallel, implement a transparent probabilistic baseline so the project does not depend entirely on one foundation model. The probability of a correct unaided response is:

$$P(Y_{iut} = 1) = \text{logistic}(\theta_{i,t} - b_u + \rho R_{i,t} + \kappa M_{i,t}) \quad (17)$$

where θ is current knowledge ability and b_u is unit difficulty. Under support level h_{iut} :

$$P(Y_{iut}^{\text{support}} = 1) = \text{logistic}(\theta_{i,t} - b_u + \rho R_{i,t} + \kappa M_{i,t} + \omega h_{iut}) \quad (18)$$

7.4 Effort and cognitive work

Define effort as a latent quantity driven by independent attempts, retrieval, generated reasoning, and time, minus answer substitution:

$$E_{iut} = \text{logistic}(a_0 + a_1 \text{Attempt}_{iut} + a_2 \text{Retrieval}_{iut} + a_3 \text{Explanation}_{iut} - a_4 \text{AnswerProvided}_{iut}) \quad (19)$$

Operational proxies should be derived from the tutor interaction: number of learner-generated steps, number of independent attempts, proportion of solution tokens generated by the learner, hint depth, and whether the final answer was revealed.

7.5 Instructional effectiveness

Instructional effectiveness depends on correctness, conceptual coverage, adaptation to the misconception, and difficulty fit:

$$F_{iut} = \text{logistic}(f_0 + f_1 \text{Correct}_u + f_2 \text{Coverage}_u + f_3 \text{Adaptation}_{iut} - f_4 \text{Mismatch}_{iut}) \quad (20)$$

7.6 Knowledge, memory, reasoning, calibration, and dependence updates

$$K_{i,t+1} = \text{clip}(K_{i,t} + \alpha_i E_{iut} F_{iut} (1 - K_{i,t}) - \delta_i K_{i,t}, 0, 1) \quad (21)$$

$$M_{i,t+1} = \text{clip}((1 - \delta_M) M_{i,t} + \eta_M \text{Retrieval}_{iut} + \eta_C \text{CorrectAfterError}_{iut}, 0, 1) \quad (22)$$

$$R_{i,t+1} = \text{clip}(R_{i,t} + \eta_R E_{iut} \text{TransferSuccess}_{iut} - \eta_O \text{Offloading}_{iut}, 0, 1) \quad (23)$$

$$C_{i,t+1} = \text{clip}(C_{i,t} + \eta_{\text{Cal}} (\text{Accuracy}_{iut} - \text{Confidence}_{iut})^2 * (-1) + \text{feedback_gain}, 0, 1) \quad (24)$$

$$D_{i,t+1} = \text{clip}(D_{i,t} + \eta_D \text{SupportUsed}_{iut} - \eta_F \text{SupportFaded}_{iut} * \text{IndependentSuccess}_{iut}, 0, 1) \quad (25)$$

The calibration equation should be implemented more transparently in code as one minus a running Brier score, rescaled to [0,1]. All update equations require parameter grids and sensitivity analysis.

7.7 Test outcomes

At fixed checkpoints, simulate tests with all support removed. Report:

- Unaided accuracy on trained items.
- Near-transfer accuracy on structurally similar items.
- Far-transfer accuracy on different surface forms and contexts.
- Retention after a specified no-practice interval.
- Calibration measured by Brier score and expected calibration error.
- Support gap: supported performance minus unsupported performance.
- Dependence: probability of requesting assistance before making an independent attempt.

$$\text{SupportGap}_{i,t} = \text{Accuracy}_{i,t}^{\text{supported}} - \text{Accuracy}_{i,t}^{\text{unaided}} \quad (26)$$

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$$\text{Brier}_{i,t} = (1/J) \sum_j (\text{Confidence}_{ij} - \text{Correct}_{ij})^2 \quad (27)$$

8. Phase IV: translate immediate predictions into longitudinal neural scenarios

8.1 Neural state representation

For every learner, maintain a functional neural state $N_{i,t,p}$ for each cortical parcel p . Initialize all states at zero or from a common standardized baseline. The state is not anatomy. It represents accumulated, model-implied functional adaptation relative to baseline.

8.2 Standardize TRIBE predictions

For each parcel, standardize immediate responses relative to the full stimulus corpus:

$$Z_{u,c,p} = (AUC_{u,c,p} - \text{mean}(AUC_{\dots,p})) / \text{sd}(AUC_{\dots,p}) \quad (28)$$

Winsorize extreme standardized values at the 1st and 99th percentiles. Retain unwinsorized results as a robustness check.

8.3 Plasticity model A: activation accumulation

$$N_{i,t+1,p} = (1 - \text{delta}_{N,p}) N_{i,t,p} + \text{eta}_{A,p} E_{iut} Z_{u,c,p} \quad (29)$$

Interpretation: repeated engaged recruitment produces cumulative specialization. This is the simplest mechanism and should not be treated as biologically definitive.

8.4 Plasticity model B: prediction-error learning

$$PE_{iut} = |Y_{iut} - P(Y_{iut}=1)| \quad (30)$$

$$N_{i,t+1,p} = (1 - \text{delta}_{N,p}) N_{i,t,p} + \text{eta}_{PE,p} PE_{iut} \text{Resolution}_{iut} Z_{u,c,p} \quad (31)$$

Interpretation: neural adaptation is strongest when a learner experiences and resolves informative error.

8.5 Plasticity model C: effort-dependent learning

$$N_{i,t+1,p} = (1 - \text{delta}_{N,p}) N_{i,t,p} + \text{eta}_{E,p} E_{iut} \text{Retrieval}_{iut} Z_{u,c,p} \quad (32)$$

Interpretation: passive exposure has weaker long-term consequences than effortful retrieval and reasoning.

8.6 Plasticity model D: hybrid

$$N_{i,t+1,p} = (1 - \delta_{N,p}) N_{i,t,p} + Z_{u,c,p} [\lambda_A E_{iut} + \lambda_{PE} PE_{iut} \text{Resolution}_{iut} + \lambda_R \text{Retrieval}_{iut} - \lambda_O \text{Offloading}_{iut}] \quad (33)$$

Constrain lambdas to be non-negative and sum to one for the positive components. The offloading penalty is estimated or varied separately. This hybrid model is the main specification because it does not force all learning through one mechanism.

8.7 Network-level neural outcomes

Aggregate parcel states to networks and derive:

- Cumulative functional recruitment by network.
- Network differentiation: between-concept distance minus within-concept distance.
- Functional concentration: share of total absolute state concentrated in the top quartile of parcels.
- Cross-network integration: mean absolute covariance of network states across episodes.
- Efficiency proxy: unaided performance divided by absolute control-network state, reported only when performance is non-trivial.
- Alignment with successful independent learning: correlation between neural state and transfer performance across learners.

$$\text{Differentiation}_c = \text{mean}[d(z_{u,z_v}) \mid \text{concept}_u \neq \text{concept}_v] - \text{mean}[d(z_{u,z_v}) \mid \text{concept}_u = \text{concept}_v] \quad (34)$$

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9. Phase V: ten-year counterfactual scenario simulation

9.1 Simulation calendar

Simulate 40 instructional weeks per year. The main exposure specification is three episodes per week, or 120 episodes per year. Run alternative specifications of one and five episodes per week. Because a fixed 120-unit curriculum would be exhausted, resample concepts with novel surface forms and progressively higher difficulty.

9.2 Core scenarios

Scenario	Personalization	Support withdrawal	Expected mechanism
Traditional	Low	Not applicable	Stable fixed instruction and persistent learner effort.
AI scaffolding, rapid fade	High	Support removed after demonstrated mastery	Personalization with strong independent retrieval.
AI scaffolding, gradual fade	High	Support declines over repeated success	Balanced personalization and autonomy.

AI scaffolding, no fade	High	Support remains available	High performance but possible dependence.
AI substitution	High	No meaningful fade	Maximum convenience and cognitive offloading.

9.3 Parameter uncertainty

Use Monte Carlo simulation with at least 2,000 parameter draws per scenario and 5,000 learners per draw when computationally feasible. If this is too expensive, use 500 parameter draws and 2,000 learners, then increase the sample for the final specification. Draw parameters from distributions rather than fixing them.

$$\Theta^b \sim p(\Theta); \text{Outcomes}^b = \text{Simulate}(\text{Scenario}, \Theta^b, \text{seed}_b) \quad (35)$$

9.4 Outcome contrasts

$$\text{ATE}_{\text{sim}}(t) = E_{b,i}[Y_{i,t}(\text{AI})] - E_{b,i}[Y_{i,t}(\text{Traditional})] \quad (36)$$

This quantity is a simulated scenario contrast, not an empirically identified average treatment effect. In the manuscript, label it SC(t), for scenario contrast, to avoid causal overstatement:

$$\text{SC}_Y(t) = \text{mean}(Y_{i,t}^{\text{AI}} - Y_{i,t}^{\text{Traditional}}) \quad (37)$$

9.5 Probability of superiority

$$\text{PrSup}_Y(t) = \text{Pr}(Y_{i,t}^{\text{AI}} > Y_{i,t}^{\text{Traditional}}) \quad (38)$$

Estimate this probability across paired learners and parameter draws. Report it together with the median difference and 90% and 95% simulation intervals.

9.6 Robustness frontier

The primary result is a frontier, not a single trajectory. Let e denote retained learner effort, a personalization quality, o answer substitution, f support fading, and eta plasticity. Define the net AI advantage:

$$G(e,a,o,f,\eta) = w_K \Delta K + w_R \Delta R + w_M \Delta M - w_D \Delta D \quad (39)$$

Classify each parameter combination as beneficial if $G > \epsilon$, neutral if $|G| \leq \epsilon$, and harmful if $G < -\epsilon$. Use equal weights in the main analysis and alternative stakeholder weights in sensitivity analyses.

9.7 Tipping points

For each key parameter, estimate the value at which the AI advantage changes sign. For example, the effort-retention tipping point is:

$$e^* = \inf\{e : G(e, a, o, f, \eta) \geq 0\} \quad (40)$$

Obtain uncertainty intervals for e^* by computing it separately in every Monte Carlo draw. Repeat for answer substitution, support fading, and forgetting.

10. Validation, falsification, and robustness

10.1 TRIBE component validation

- Reproduce at least one official TRIBE demonstration before using the model.
- Use public held-out naturalistic fMRI data, when licensing and formatting permit, to test whether the pipeline retains predictive validity outside the official examples.
- Confirm that predicted responses change sensibly when text is temporally shuffled or replaced by irrelevant content.
- Run identical stimuli twice and verify deterministic equality under fixed settings.
- Report model failures, unsupported modalities, truncated inputs, and missing regions.

10.2 Synthetic learner validation

- Check monotonicity: higher prior knowledge should not reduce baseline accuracy on average.
- Check difficulty: higher item difficulty should reduce unaided accuracy.
- Check support: assistance should increase immediate supported accuracy.
- Check forgetting: longer no-practice intervals should reduce retention.
- Check fading: successful independent performance should reduce dependence under fading policies.
- Compare Centaur or Minitaur trajectories with the transparent probabilistic baseline.

10.3 Negative controls

- Incorrect but fluent explanations.
- Sentence-shuffled explanations with equal words and duration.
- Irrelevant explanations matched in style and length.
- Randomly permuted TRIBE predictions across units.
- Randomly permuted condition labels within matched unit sets.
- Zero-plasticity scenario, which must produce no long-term neural divergence.
- Zero-effort-sensitivity scenario, which should eliminate effort-mediated differences.

10.4 Specification curve

Run a structured multiverse over all defensible choices: reading speed, parcellation, aggregation weights, TRIBE metric, learner engine, forgetting function, effort function, plasticity mechanism, exposure rate, and outcome

weights. Rank specifications by theoretical plausibility before examining results. Plot every estimated year-10 contrast with its uncertainty interval.

10.5 Variance and uncertainty decomposition

Estimate how much uncertainty in the year-10 outcome is attributable to stimulus generation, learner heterogeneity, behavioural-model stochasticity, plasticity parameters, and scenario assumptions. A practical approach is a random-effects ANOVA or Sobol sensitivity analysis:

$$Var(Y) = V_{stimulus} + V_{learner} + V_{behavior} + V_{plasticity} + V_{scenario} + V_{residual} \quad (41)$$

The final conclusions should focus on factors that explain large shares of output uncertainty.

10.6 Falsification criteria

The project should explicitly accept null or indeterminate conclusions. We will not claim a meaningful difference if any of the following occurs:

- Condition effects disappear after matching content and duration.
- TRIBE contrasts are smaller than variation across harmless stylistic regenerations.
- Long-term direction reverses across equally plausible plasticity models.
- The year-10 result is driven almost entirely by arbitrary parameter bounds.
- Negative controls generate effects of similar size to the substantive conditions.
- The simulation cannot distinguish scaffolding from substitution after support is removed.

11. Statistical analysis plan

11.1 Corpus balance

- Report means, standard deviations, and SMDs for every matching feature.
- Plot paired traditional-AI differences by unit.
- Use equivalence tests only as descriptive diagnostics, not proof of identical materials.

11.2 Immediate cortical contrasts

Fit paired hierarchical models for parcel or network summaries:

$$y_{ucp} = \alpha_u + \alpha_p + \beta_1 Scaffold_c + \beta_2 Substitute_c + \beta_3 Difficulty_u + \beta_4 Domain_u + \epsilon_{ucp} \quad (42)$$

If using Bayesian estimation, place regularizing priors on condition effects and hierarchical parcel effects. Correct for multiple comparisons when presenting parcel-level maps. Network-level outcomes should be preregistered as primary.

11.3 Learning trajectories

Model simulated unaided performance with a generalized additive mixed model or hierarchical logistic model:

$$\text{logit } P(Y_{i,t}=1) = \alpha_i + f_c(t) + \beta_1 \text{Prior}_i + \beta_2 \text{Domain} + \beta_3 \text{Difficulty} + \text{interactions} \quad (43)$$

Because the data are simulated, p-values are not intrinsically meaningful. Emphasize effect magnitude, uncertainty induced by the simulation design, and robustness across parameter draws.

11.4 Neural trajectory contrasts

For each network n and checkpoint t , report paired scenario contrasts, standardized effects, and probability of superiority:

$$d_{n,t} = \text{mean}(N_{i,t,n}^{AI} - N_{i,t,n}^T) / \text{sd}(N_{i,t,n}^{AI} - N_{i,t,n}^T) \quad (44)$$

11.5 Mediation as mechanism decomposition

Do not call this causal mediation. Use computational decomposition to quantify how much of the scenario contrast travels through effort, instructional effectiveness, support dependence, and predicted neural engagement. Re-run the simulation while holding each mediator at the traditional-condition distribution. The reduction in the contrast is the mechanism contribution under the model.

12. Execution plan

Stage	Student 1	Student 2	Joint checkpoint
1. Setup	Repository, curriculum schema, validation functions	TRIBE installation and official examples	Environment and reproducibility audit
2. Corpus	Reference units and three conditions	Timing conversion and TRIBE input adapters	Corpus matching approval
3. Immediate predictions	Support with content diagnostics	Run TRIBE, aggregate parcels and networks	First condition-contrast report
4. Learner engine	Transparent baseline and Minitaur/Centaur integration	Neural-state data structures	Unit tests and sanity checks
5. Longitudinal model	Learning, retention, transfer, dependence	Plasticity models and neural trajectories	One-year pilot simulation
6. Full scenarios	Behavioural results	Ten-year Monte Carlo and phase diagrams	Robustness frontier
7. Validation	Learner negative controls	TRIBE and plasticity negative controls	Specification curve and uncertainty report
8. Finalization	Methods and behavioural findings	Neural and simulation findings	Paper, appendix, dashboard, presentation

12.1 Decision gates

16. Do not process the full corpus until 10 units pass all matching and correctness tests.
17. Do not launch full TRIBE inference until official examples are reproduced.
18. Do not run ten-year simulations until the one-year model passes monotonicity and zero-effect tests.
19. Do not interpret a cortical metric until its mathematical meaning and direction are documented.
20. Do not write substantive conclusions until the specification curve and negative controls are complete.

13. Required figures and tables

- Figure 1. Conceptual pipeline from educational material to TRIBE response, learner update, plasticity update, and long-term scenario.
- Figure 2. Corpus matching diagnostics across conditions.
- Figure 3. Predicted network-level responses for matched traditional, scaffolding, and substitution stimuli.
- Figure 4. Representational similarity matrices by instructional condition.
- Figure 5. One-year knowledge, transfer, and dependence trajectories.
- Figure 6. Year-10 scenario distributions with uncertainty intervals.
- Figure 7. Phase diagram of personalization, effort retention, and answer substitution.
- Figure 8. Specification curve across all defensible modelling choices.
- Table 1. Instructional conditions and fixed versus varying features.
- Table 2. Model components, inputs, outputs, assumptions, and validation tests.
- Table 3. Main parameter distributions and sensitivity ranges.
- Table 4. Immediate cortical contrasts.
- Table 5. One-, five-, and ten-year scenario contrasts.
- Table 6. Negative controls and robustness outcomes.

14. Main risks and mitigation

Risk	Why it matters	Mitigation
TRIBE is not an educational plasticity model	Immediate response may be mistaken for learning	Separate immediate encoding from longitudinal updates and label outputs precisely.
Synthetic learner lacks educational validity	Human-like choices do not guarantee realistic learning	Use transparent baseline, multiple engines, sanity tests, and public benchmarks where possible.
Long-term result is assumption-driven	Ten years magnifies small assumptions	Use broad priors, tipping points, specification curves, and uncertainty decomposition.
Conditions differ in uncontrolled ways	Effects may reflect style or duration	Strict matching, static-AI control, paraphrase controls, and paired analyses.

More activation is interpreted as better	Neural magnitude is ambiguous	Focus on representational structure and report direction-neutral metrics.
Computational cost becomes excessive	Full vertex outputs and Monte Carlo runs are large	Pilot on parcels and subsets, cache outputs, vectorize simulations, scale only final models.
AI-generated factual errors	Invalid material contaminates all later stages	Deterministic validators, independent judging, and mandatory manual review of failures.

15. How the final paper must be written

Language rule: Use “predicted cortical response,” “simulated learner,” “model-implied trajectory,” and “scenario contrast.” Avoid “the brain learns,” “AI causes,” “we demonstrate long-term effects,” and “digital twin of the student.”

The paper should be structured as a methods and scenario-analysis contribution. The introduction should motivate uncertainty about the cognitive consequences of AI-based education and the impracticality of waiting a decade for evidence. The theoretical section should distinguish adaptive scaffolding from cognitive substitution. The methods should foreground matched stimuli, component validation, transparent update equations, and uncertainty propagation. The discussion should identify which assumptions drive divergent futures and specify the empirical measurements that future research must collect.

15.1 Minimum claims for a successful project

- We built and validated a reproducible computational framework.
- We identified measurable differences or credible null differences in model-predicted immediate responses.
- We showed how educational futures depend on explicit combinations of personalization, effort, support withdrawal, and plasticity.
- We identified robust regions of the parameter space and regions where conclusions remain indeterminate.
- We translated the model into concrete empirical priorities for future educational neuroscience.

16. Core references and resources

Binz, M., Akata, E., Bethge, M., Brändle, F., Callaway, F., Coda-Forno, J., Dayan, P., Demircan, C., Eckstein, M. K., Gureckis, T. M., Mattar, M. G., and Schulz, E. (2025). A foundation model to predict and capture human cognition. *Nature*. <https://doi.org/10.1038/s41586-025-09215-4>

Wang, E. Y., Fahey, P. G., Ding, Z., and colleagues. (2025). Foundation model of neural activity predicts response to new stimulus types. *Nature*, 640, 470-477. <https://doi.org/10.1038/s41586-025-08829-y>

Banville, H. J., King, J. R., Raugel, J., Rapin, J., Begany, K., d Ascoli, S., Brooks, T., and Benchetrit, Y. (2026). A foundation model of vision, audition, and language for in-silico neuroscience. *Meta AI Research*. <https://ai.meta.com/research/publications/a-foundation-model-of-vision-audition-and-language-for-in-silico-neuroscience/>

Official TRIBE v2 repository and released checkpoints. <https://github.com/facebookresearch/tribev2>

Official Centaur/Psych-101 code and model resources linked from the *Nature* article and authors repositories.

Appendix A. Final execution checklist

- ☐ Repository builds from a clean environment.
- ☐ All 120 units have deterministic answer validation.
- ☐ Three conditions are factually equivalent and matched on observable features.
- ☐ Scaffolding and substitution policies pass leakage and compliance tests.
- ☐ Official TRIBE example is reproduced.
- ☐ All TRIBE outputs are cached with metadata and timestamps.
- ☐ Parcel and network mappings are documented.
- ☐ Synthetic learner monotonicity tests pass.
- ☐ Zero-plasticity and shuffled-condition controls produce null results.
- ☐ One-year pilot behaves sensibly before the ten-year run.
- ☐ Every parameter has a source, rationale, or explicit “assumption” label.
- ☐ All main conclusions survive the preregistered robustness set or are reported as indeterminate.
- ☐ No result is described as causal or as an observed future brain state.
- ☐ All figures and tables regenerate from scripts.
- ☐ Final README enables another team to reproduce the pipeline.